



Cairo University – Faculty of Engineering
Electronics and Electrical Communications
Mainstream – Second year
Fields and Wave Propagation - EECG252



Assignment 1

Submitted for EECG252 Assignment

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1 ANALYTICAL SOLUTION

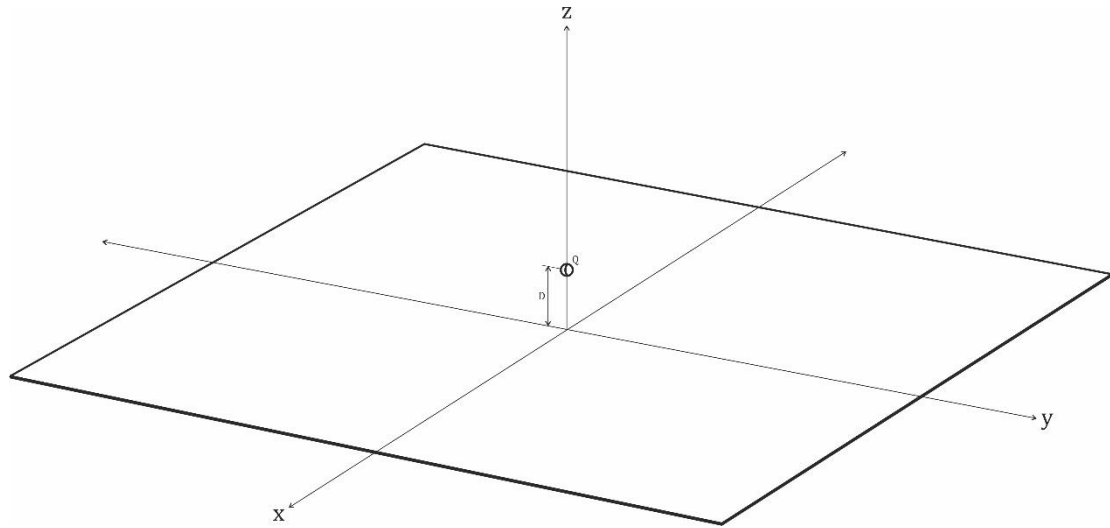


Figure 1.1: The problem illustration.

Assuming $Q = 2 \text{ nC}$, $D = 5 \text{ m}$.

1.1 HAND ANALYSIS

1.1.1 Finding Electrical Potential Everywhere

We will start the hand analysis by performing the method of images:

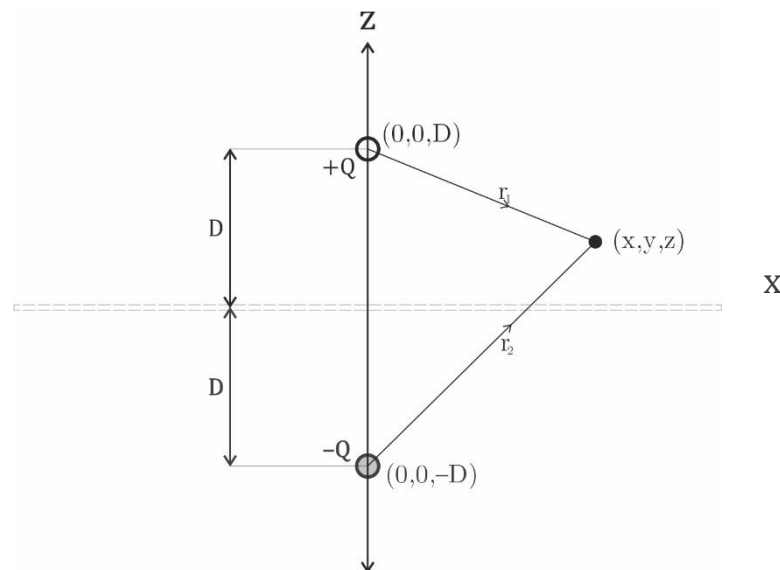


Figure 1.2: Method of Images.

We know that the Electrical Field of point charge (Q) at any point is:

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \quad (1.01)$$

For the main point charge (Q) Electrical Field is:

$$\underline{E}_1 = \frac{Q}{4\pi\epsilon_0 r_1^2} \hat{r}_1 \quad (1.02)$$

Where:

$$|r_1| = \sqrt{x^2 + y^2 + (z - D)^2} \quad (1.03)$$

For the image point charge (-Q) Electrical Field is:

$$\underline{E}_2 = \frac{-Q}{4\pi\epsilon_0 r_2^2} \hat{r}_2 \quad (1.04)$$

Where:

$$|r_2| = \sqrt{x^2 + y^2 + (z + D)^2} \quad (1.05)$$

To get total Electrical Field:

$$\vec{E}_T = \vec{E}_1 + \vec{E}_2 \quad (1.06)$$

$$\underline{E}_T = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right) \hat{E} \quad (1.07)$$

We know that:

$$\varphi = - \int \underline{E}_T \cdot d\vec{l} \quad (1.08)$$

$$\varphi = - \int \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right) dr \quad (1.09)$$

$$\varphi = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (1.10)$$

By substituting we get potential everywhere:

$$\varphi = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{x^2 + y^2 + (z - D)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z + D)^2}} \right) \quad (1.11)$$

Getting values of φ at different points:

- At (0, 0, 0):

$$\varphi = 0 \text{ V} \quad (1.12)$$

- At (-10, -10, 5):

$$\varphi = 0.2332386743 \text{ V} \quad (1.13)$$

- At (7, 7, 7):

$$\varphi = 0.624317043 \text{ V} \quad (1.14)$$

1.1.2 Finding surface charge density on the infinite plane

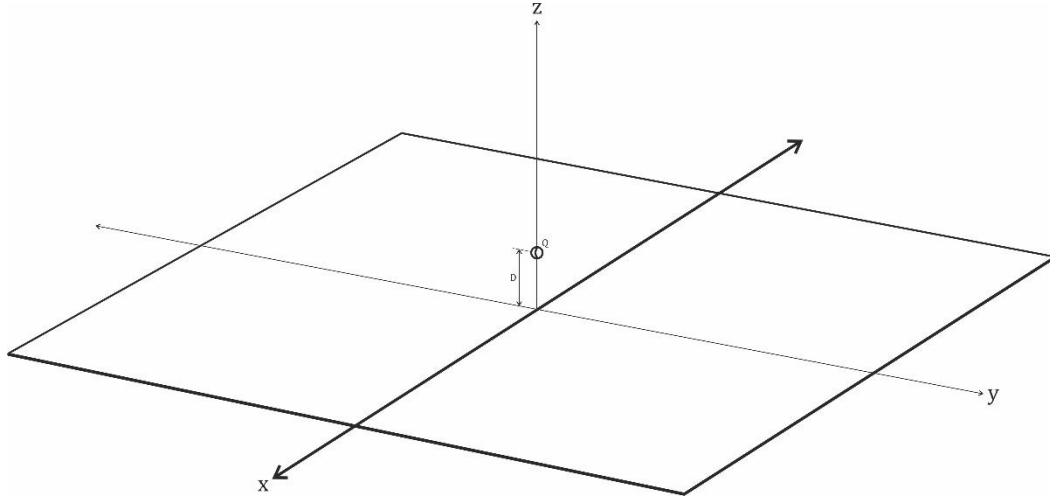


Figure 1.3: Added an infinite straight line below the point charge to calculate surface charge density on it.

By using Gauss's law:

$$\oiint_S \underline{E}_T \cdot d\underline{S} = \frac{\sum Q_{enclosed}}{\epsilon_0} \quad (1.15)$$

By Knowing:

$$Q_{enclosed} = \sigma \times S \quad (1.16)$$

Where σ is the surface charge density.

We will only take the Z-component of the electrical field as it is the only component

parallel to the area vector the other components are perpendicular to, they are zeroed by the dot product:

$$\underline{E}_{z_T} \times S = \frac{\sigma \times S}{\epsilon_0} \quad (1.17)$$

$$\sigma = \epsilon_0 \underline{E}_{z_T} \quad (1.18)$$

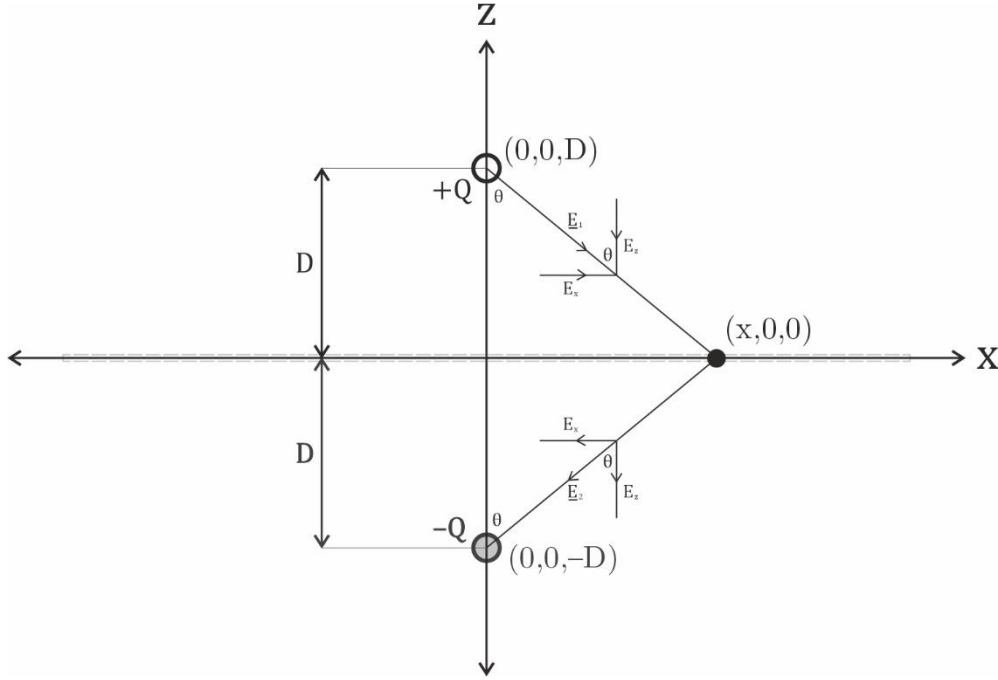


Figure 1.4: Getting the Z component of E_1 and E_2 .

Getting that:

$$\underline{E}_{z_1} = |E_1| \cos(\theta) (-\hat{z}) \quad (1.19)$$

$$\underline{E}_{z_2} = |E_2| \cos(\theta) (-\hat{z}) \quad (1.20)$$

Where:

$$\cos(\theta) = \frac{D}{|r|} \quad (1.21)$$

By substituting in equation (1.18) we get from equations (1.21), (1.19) and (1.07):

$$\sigma = \epsilon_0 \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{|r_1|^2} + \frac{|-Q|}{|r_2|^2} \right) \frac{D}{|r|} (-\hat{z}) \quad (1.22)$$

As $y=0$, $z=0$ on x-axis so:

$$|r_1| = |r_2| = |r| = \sqrt{x^2 + D^2} \quad (1.23)$$

By updating equation (1.22):

$$\sigma = \frac{-QD}{2\pi|r|^3} \quad (1.24)$$

$$\sigma = \frac{-QD}{2\pi\sqrt{(x^2 + D^2)^3}} \quad (1.25)$$

Getting values of σ at different points:

- At (0, 0, 0):

$$\sigma = -1.273239545 \times 10^{-11} \text{ C/m}^2 \quad (1.26)$$

- At (-5, 0, 0):

$$\sigma = -4.501581581 \times 10^{-12} \text{ C/m}^2 \quad (1.27)$$

- At (10, 0, 0):

$$\sigma = -1.138820069 \times 10^{-12} \text{ C/m}^2 \quad (1.28)$$

1.2 MATLAB RESULTS

Listing 1.1: MATLAB code for calculating the Electric Potential at certain points.

```

1  % Auther : Mina Wael Tanagho Fahmy Sedrak
2  % Section : 4
3  % BN      : 13
4  % ID      : 91240804
5
6  clc;
7  clear all;
8  close all;
9
10 %% Calculations
11
12 Q = 2e-9;
13 D = 5;
14 e0 = 8.854e-12;
15
16 X = [0,-10,7];
17 Y = [0,-10,7];
18 Z = [0,5,7];
19 V = zeros(1,3);
20
21 for i = 1:3
22
23     V(i) = ...
24     (Q/(4*pi*e0))*((1./((sqrt(X(i).^2 + Y(i).^2 + (Z(i) - D).^2)))-...
25     (1./((sqrt(X(i).^2 + Y(i).^2 + (Z(i) + D).^2))))
26 end

```

```

21 - V(i) = ...
22 - (Q/(4*pi*e0))*((1./sqrt(X(i).^2 + Y(i).^2 + (Z(i) - D).^2))-...
23 - (1./sqrt(X(i).^2 + Y(i).^2 + (Z(i) + D).^2)));
24 - end
25
26 - display(V);

```

Command Window

```

V =
    0    0.2332    0.6243
fx >>

```

Figure 1.5: MATLAB results part_1.

We can compare between the hand analysis at equations (1.12), (1.13) and (1.14) and the MATLAB results, they agree with each other.

Listing 1.2: MATLAB code for plotting the surface charge density and marking certain values.

```

1  % Auther : Mina Wael Tanagho Fahmy Sedrak
2  % Section : 4
3  % BN      : 13
4  % ID      : 91240804
5
6  %% plot code
7  clc;
8  clear all;
9  close all;
10
11  Q = 2e-9;
12  D = 5.04;
13
14  x_axis = -100:1:100;
15
16  surface_charge_density = (-Q * D) ./ (2 * pi * (x_axis.^2 + D^2).^(1.5));
17
18  figure();
19  plot(x_axis, surface_charge_density, 'LineWidth', 2);
20  grid on;
21
22  x_point = [0, -5, 10];
23  y_point = (-Q * D) ./ (2 * pi * (x_point.^2 + D^2).^(1.5));
24  hold on;
25
26  scatter(x_point, y_point, 'filled');
27
28  for i = 1:3
29      text(x_point(i), y_point(i), sprintf(' (%.f, %.4e)', x_point(i),
30      y_point(i)), 'FontWeight', 'bold');
31  end
32
33  xlabel("x-axis");
34  ylabel("Surface Charge Density (C/m^2)");

```

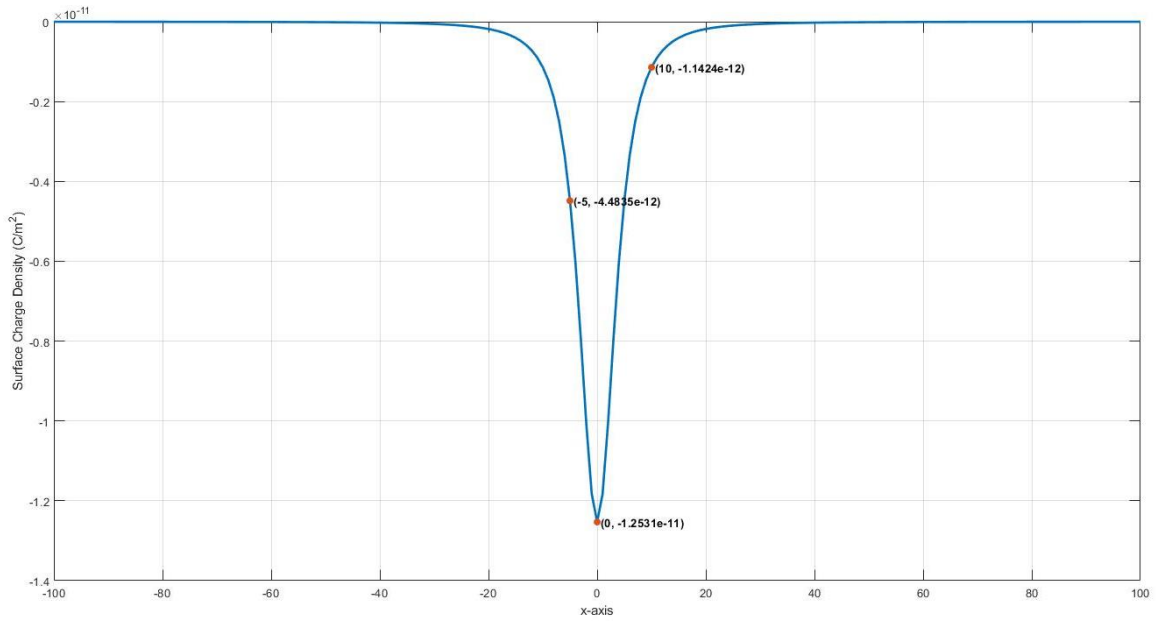


Figure 1.6: MATLAB results part_2.

We can compare between the hand analysis at equations (1.26), (1.27) and (1.28) and the MATLAB results, they agree with each other.

2 CST SOLUTION

2.1 SETTING UP

Starting with placing a sphere with radius equals 0.05 m as a point charge having a charge Q equals $2e-9$ C and has a material type of PEC at a distance D equals 5 m from the x-y plane.

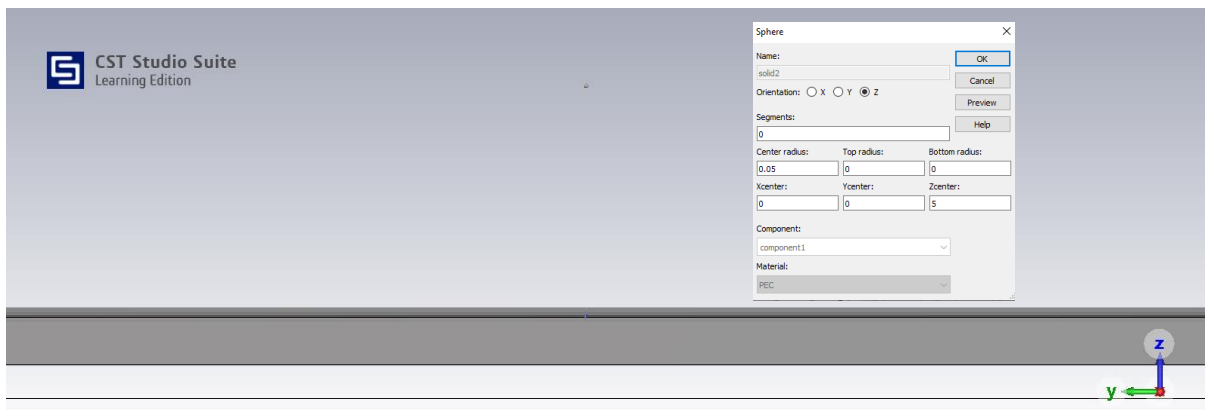


Figure 2.1: Placing the point charge.

Then placed the infinite surface from -100 to +100 in the x direction and from -100 to +100 in the y direction and thickness of 1 m, with a material type of PEC.

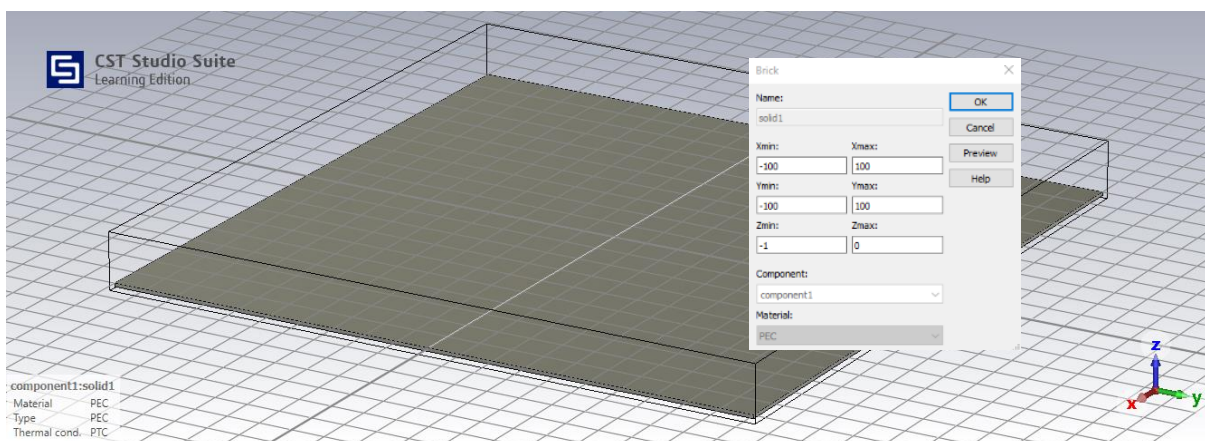


Figure 2.2: Placing the infinite plate.

Then added the straight line for the calculation of the surface charge density at the x-axis from -100 to +100.

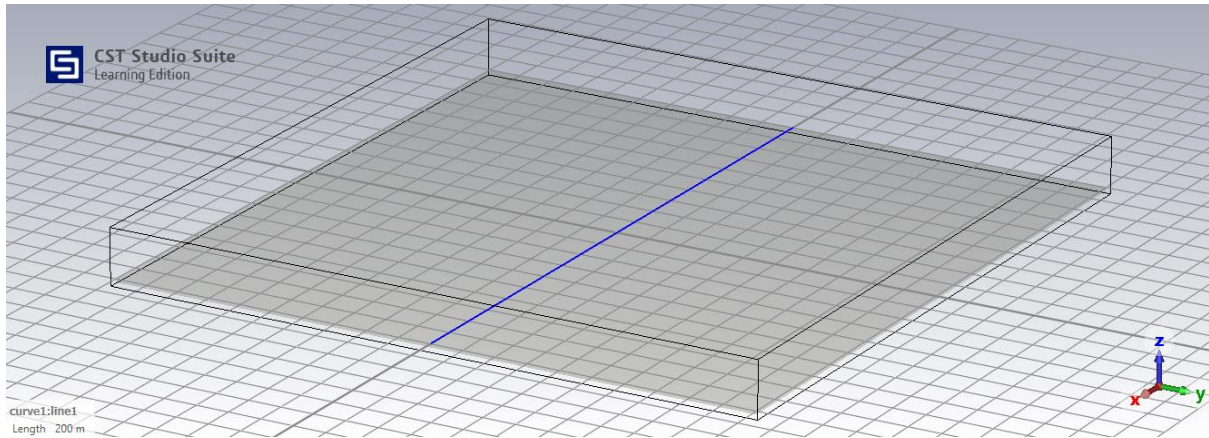


Figure 2.3: Placing the infinite straight line.

2.2 Finding Electrical Potential Everywhere

By running the Set up solver and going to the 2D/3D Results we get the potential everywhere as follows in **Figure 2.4** and **Figure 2.5**

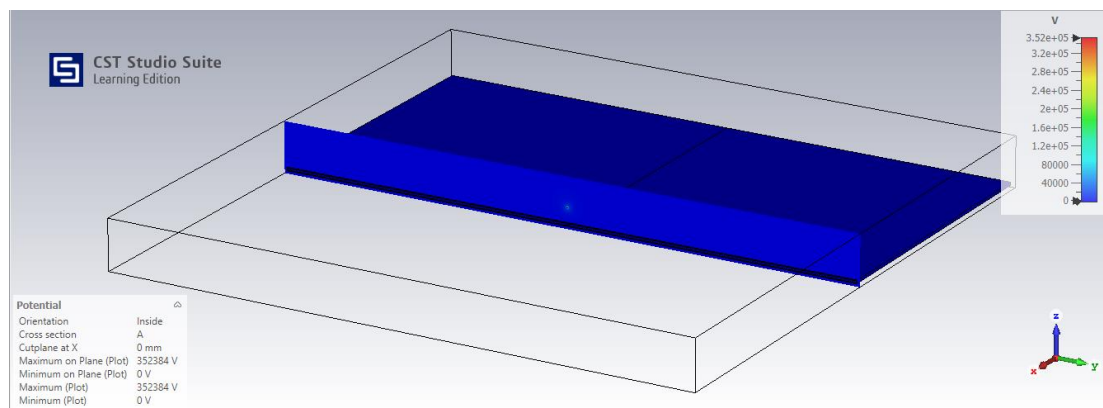


Figure 2.4: Potential Everywhere.

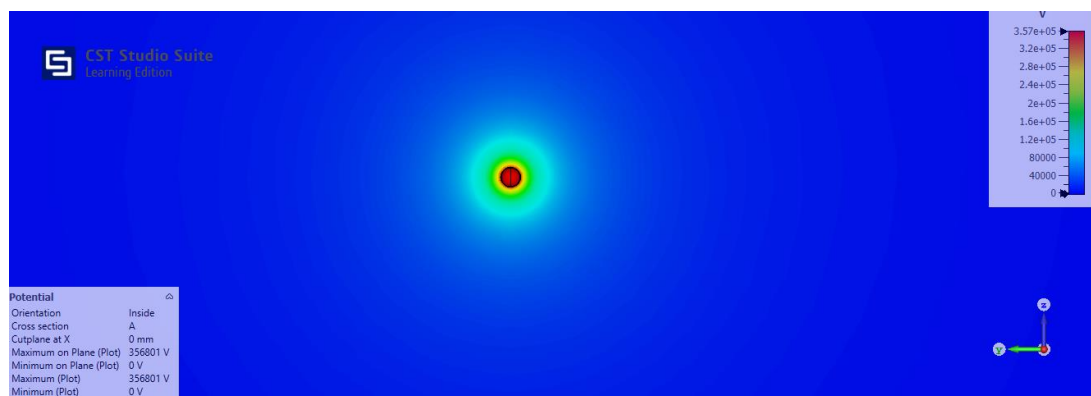


Figure 2.5: Potential Everywhere Section view.

Then got to the Post-Processing Result Templates and calculated potential at certain points $(0, 0, 0)$, $(-10, -10, 5)$, $(7, 7, 7)$.

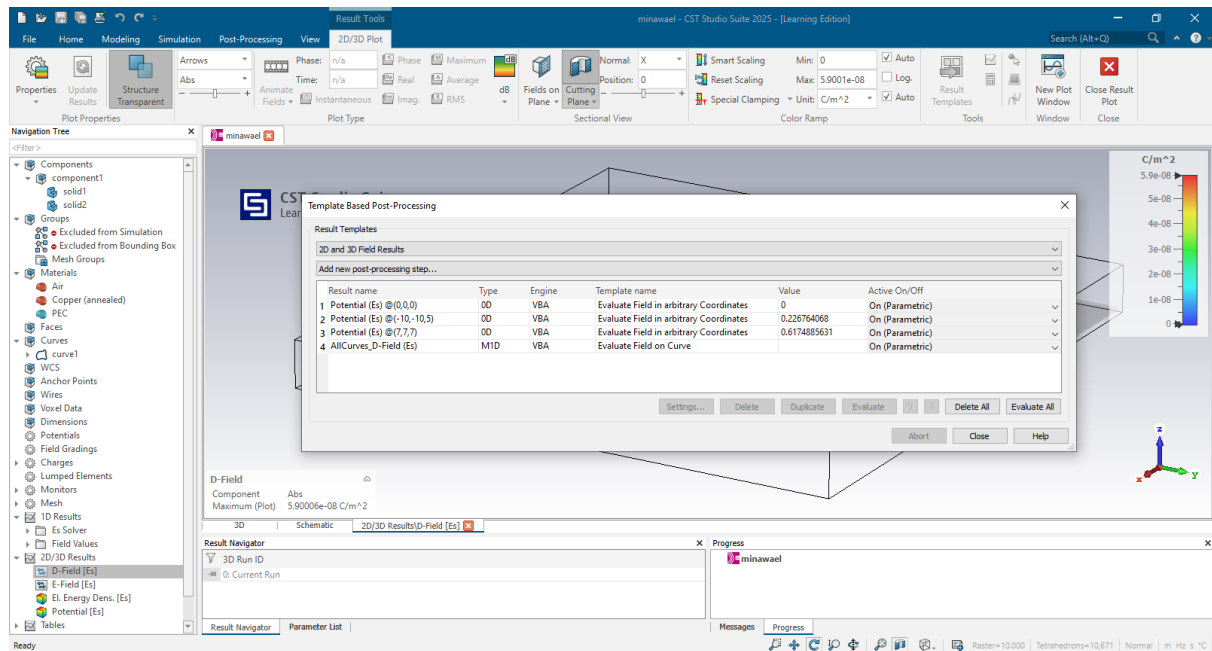


Figure 2.6: Result Templates.

Table 2.1: This table showed *Result Templates*

Result name	Value
Potential (E_s) @ $(0, 0, 0)$	0
Potential (E_s) @ $(-10, -10, 5)$	0.226764068
Potential (E_s) @ $(7, 7, 7)$	0.6174885631

As we can see the results are so close as in the hand analysis and MATLAB results before.

2.3 FINDING SURFACE CHARGE DENSITY ON THE INFINITE PLANE

After placing the infinite straight line, we got to the Post-Processing Result Templates and plotted the surface charge density graph against the length in m.

As in **Figure 2.7**, then added some value markers to compare with hand analysis and MATLAB results. As in **Figure 2.8**.

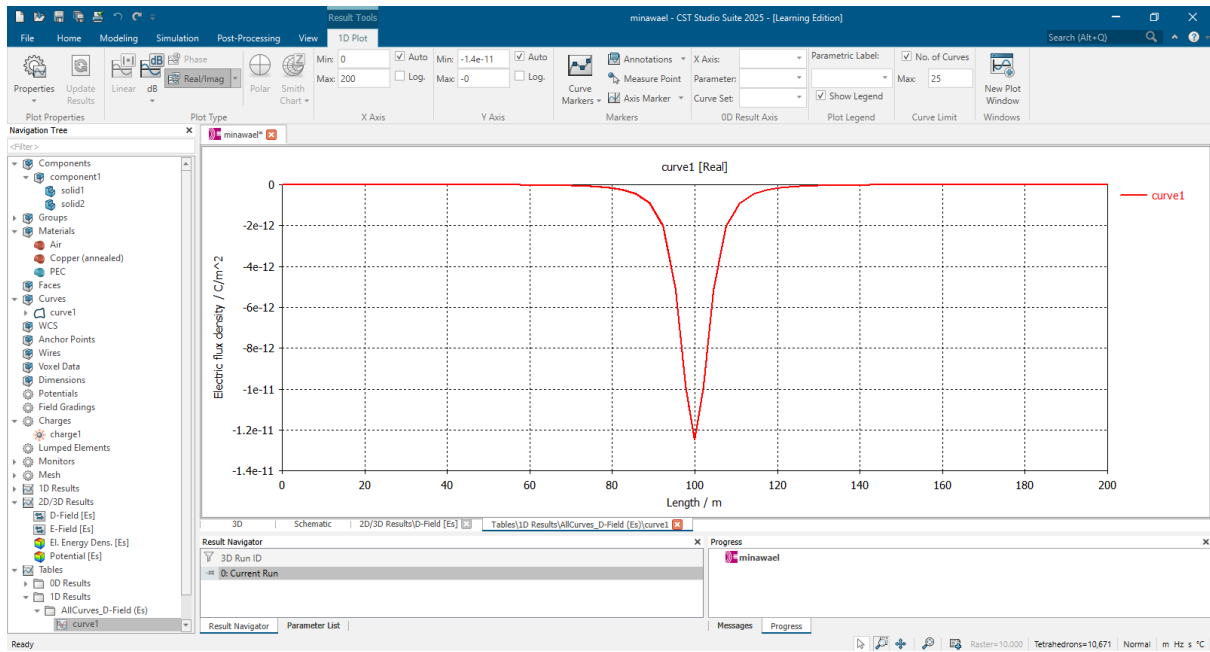


Figure 2.7: Plot of surface charge density vs length in m.

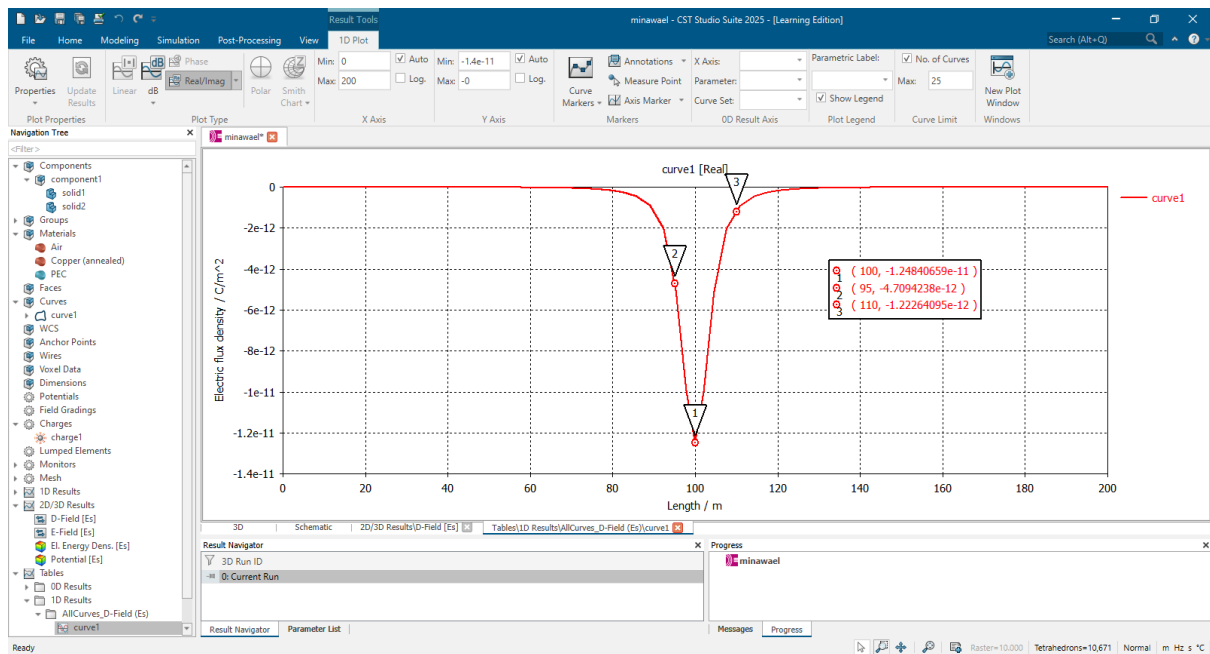


Figure 2.8: Marking the values we are interested in and enabling the legend.

As we can see the plot is the same as the plot of MATLAB.

Also, by comparing values from Hand analysis equations at (1.26), (1.27) and (1.28) we can say that they are really so close.

3 COMPARISON BETWEEN RESULTS

Calculating the error in one of the results of potential at point (-10, -10, 5) from CST results and hand analysis.

$$Error (\%) = \frac{|0.226764068 - 0.2332386743|}{0.226764068} \times 100 = 2.855\% \quad (3.1)$$

Calculating the error in one of the results of surface charge density at point (-5, 0, 0) from CST results and hand analysis.

$$Error (\%) = \frac{|-4.7094238 \times 10^{-12} - (-4.501581581 \times 10^{-12})|}{-4.7094238 \times 10^{-12}} \times 100 = 4.413\% \quad (3.2)$$

Both errors are very small. So, they are acceptable.

The reason why this error happens is that:

- Math (Hand Analysis): Infinite, continuous, and perfect.
- CST: Finite, meshed, and limited by the size of the box.

- Note:

The used plate at CST is not infinite it has an area of 40,000 m² which is 200 × 200 in meters.